Triangles & Congruence

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

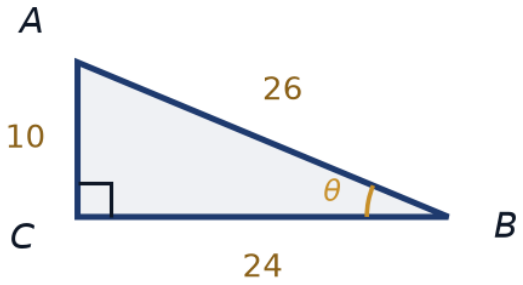
A data set recorded for review: equations of lines consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(3)$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Triangles & Congruence · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a 45°–45°–90° right triangle, the hypotenuse has length $10\sqrt{2}$ cm. What is the length of each leg, and what is the area of the triangle?

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Right Triangles & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right ΔABC with right angle at C, altitude CD is drawn to hypotenuse AB. If $AD = 4$ and $DB = 9$, find the length of leg BC.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Points A, B, C lie on a circle with center O. Inscribed angle $\angle ABC$ intercepts minor arc AC and has measure 38°. What is the measure of the central angle $\angle AOC$ intercepting the same arc?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Similarity & Proportional Reasoning · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Circles · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Foundations & Reasoning · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Polygons & Quadrilaterals · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

[Area, Surface Area & Volume · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Transformations & Symmetry

[Transformations & Symmetry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Foundations & Reasoning

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Coordinate Geometry

A bivariate regression model for student preparation hours x and exam grade y yields the line of best fit $\hat{y} = 4.8x + 52$. A student who studied for 8 hours scored 94 on the exam. (a) What is the model's predicted score for this student? (b) Compute and interpret the student's regression residual.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability

Section I of EST I Math contains 20 questions in 25 minutes. If a student answers the first 12 questions in an average of 50 seconds each, how many seconds per question are available for the remaining 8 questions?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Lines & Angles

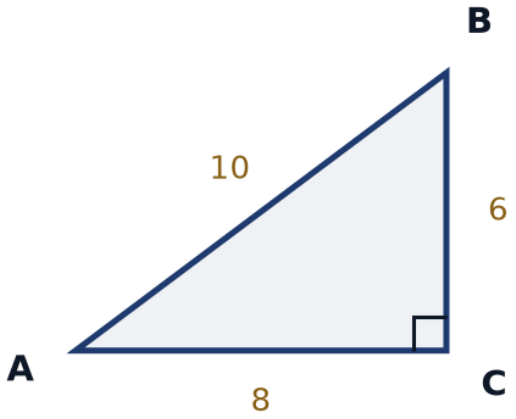
A student has scores of 78, 85, 92, and 81 on four exams. What score must the student earn on the fifth exam to achieve an overall arithmetic mean of 85 across all five exams?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 93°

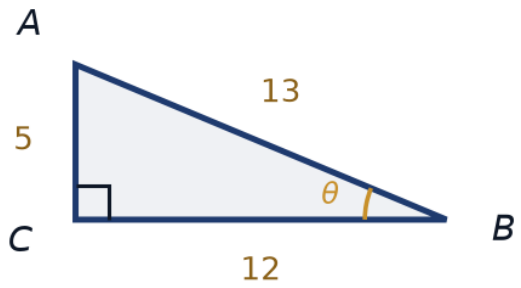
b. 103°

c. 85°

d. 138°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 92°

b. 102°

c. 84°

d. 137°

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 5$ in extension: perpendiculars and distance, evaluate $f(3)$.

a. 14

b. 17

c. 9

d. 18

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a triangle, two angles measure 14° and 101°. What is the measure of the third angle?



a. 101°

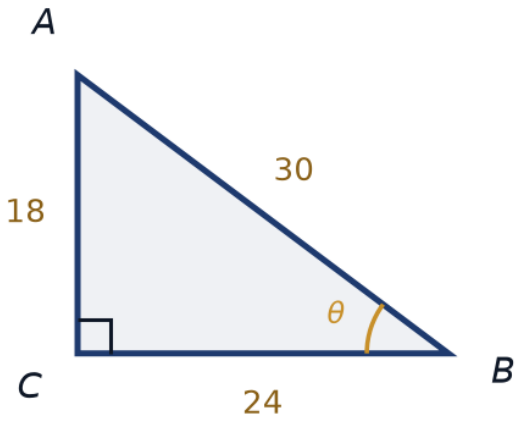
b. 65°

c. 115°

d. 14°

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An isosceles triangle has a vertex angle of 12°. What is the measure of each base angle?



a. 84°

b. 6°

c. 12°

d. 168°

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC and ΔXYZ, we are given that $AB = XY$, $BC = YZ$, and $\angle B = \angle Y$. Which of the following additional conditions would NOT be sufficient to prove that $\Delta ABC \cong \Delta XYZ$?

a. $\angle A = \angle X$

b. $AC = XZ$

c. $\angle C = \angle Z$

d. $AB = XZ$

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, medians $\overline{AD}$ and $\overline{BE}$ intersect at the centroid G. The lengths of the medians are $AD = 12$ and $BE = 9$. Which of the following represents the complete range of all possible lengths for side $\overline{BC}$?

a. $4 < BC < 20$

b. $2 < BC < 14$

c. $3 < BC < 21$

d. $2 < BC < 10$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangles ABC and DEF, it is given that $AB \cong DE$ and $\angle B \cong \angle E$. Which of the following additional conditions is NOT sufficient by itself to guarantee that $\Delta ABC \cong \Delta DEF$?

a. $AC \cong DF$

b. $BC \cong EF$

c. $\angle A \cong \angle D$

d. $\angle C \cong \angle F$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Can segments of length 10, 12, and 12 form a triangle?

a. Only if the triangle is isosceles

b. No, they cannot form a triangle

c. Yes, they can form a triangle

d. Only if the triangle is right

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two triangles have two pairs of corresponding angles and a non-included side are congruent. Which congruence postulate proves the triangles congruent? (variant 5)

a. ASA

b. SAS

c. SSS

d. AAS

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two triangles have two pairs of corresponding angles and the included side are congruent. Which congruence postulate proves the triangles congruent? (variant 14)

a. SSS

b. ASA

c. SAS

d. AAS

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two triangles have two pairs of corresponding sides and the included angle are congruent. Which congruence postulate proves the triangles congruent? (variant 2)

a. AAS

b. ASA

c. SSS

d. SAS

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two triangles have all three pairs of corresponding sides are congruent. Which congruence postulate proves the triangles congruent? (variant 2)

a. AAS

b. SAS

c. SSS

d. ASA

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Triangles $ABC \cong DEF$ by SAS. If $\angle A = (3x + 7)^\circ$ and the corresponding $\angle D = (1x + 13)^\circ$, find $\angle A$.

a. 16°

b. 24°

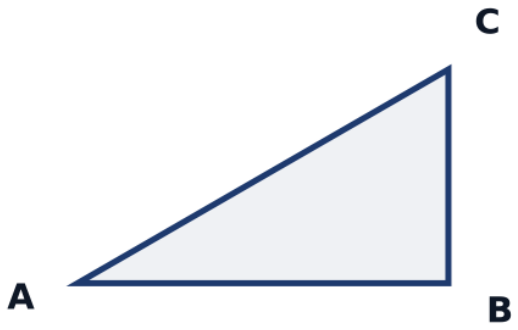
c. 3°

d. 164°

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35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Triangle with $\angle A = 30^\circ$, $\angle B = 90^\circ$, side $a = 5$ (opposite A). Find side b (Law of Sines).



a. 10

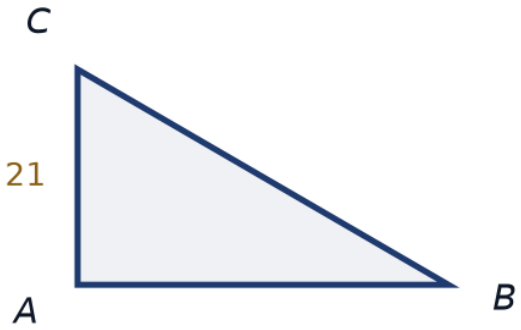
b. 5

c. 12

d. $\frac{5}{2}$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A 30-60-90 right triangle has a short leg of length 21. What is the length of the longer leg?



a. $21\sqrt{3}$

b. 21

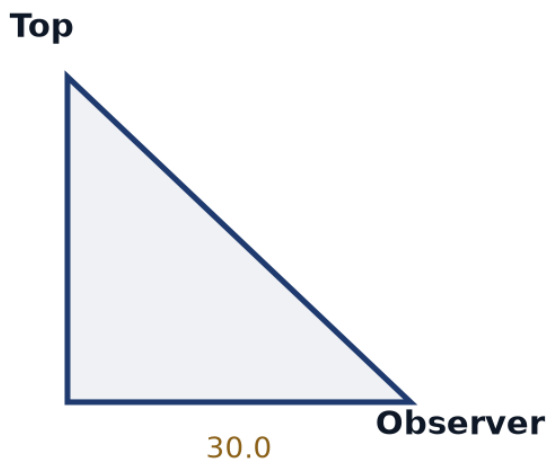
c. 42

d. $21\sqrt{2}$

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

From a point 30 m from the base of a tower, the angle of elevation to the top is 60°. Find the height of the tower, rounded to two decimal places.



a. 54.96

b. 30

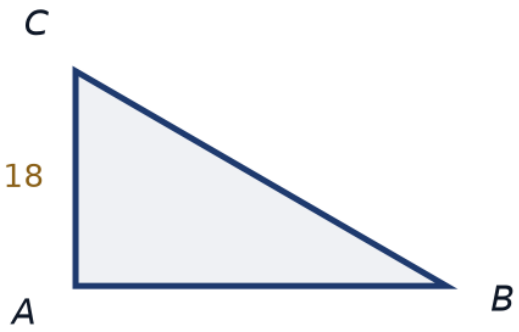
c. 51.96

d. 25.98

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Coordinate Geometry

A 30-60-90 right triangle has a short leg of length 18. What is the length of the hypotenuse?



a. 18

b. $18\sqrt{2}$

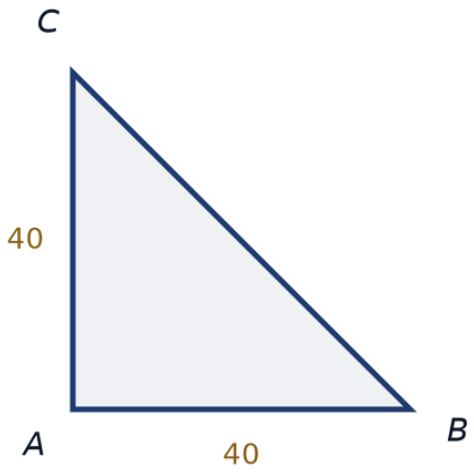
c. $18\sqrt{3}$

d. 36

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability

A 45-45-90 right triangle has legs of length 40. What is the length of the hypotenuse?



a. $40\sqrt{3}$

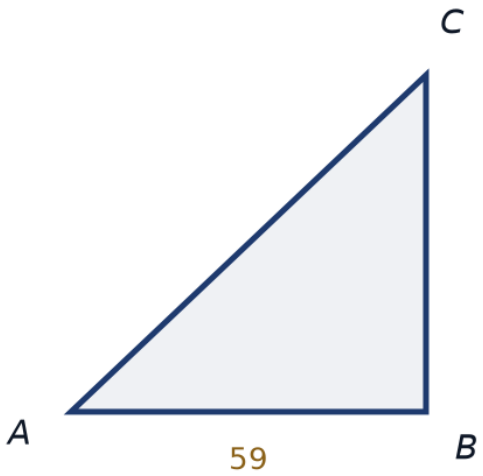
b. $40\sqrt{2}$

c. 40

d. 80

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Lines & Angles

In a right triangle, one acute angle measures 45° and the side adjacent to it is 59. Find the length of the side opposite the 45° angle, rounded to two decimal places.



a. 60.5

b. 57.0

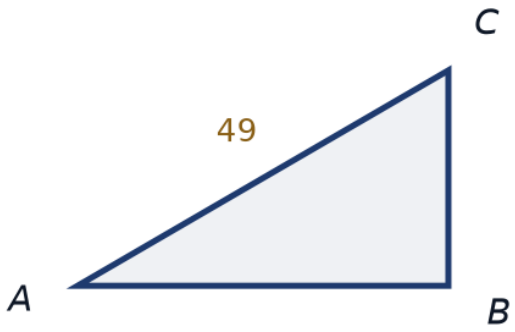
c. 59.0

d. 70.8

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In a right triangle, one acute angle measures 30° and the hypotenuse is 49. Find the length of the side adjacent to the 30° angle, rounded to two decimal places.



a. 40.43

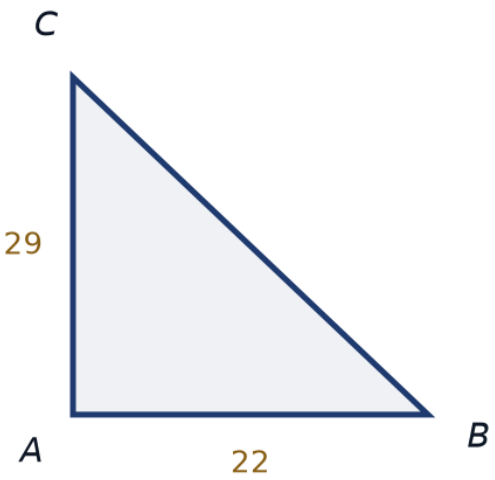
b. 43.93

c. 42.43

d. 50.92

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A right triangle has legs of length 22 and 29. What is the length of the hypotenuse? (Simplify if possible.)



a. $5\sqrt{53}$

b. $\sqrt{1321}$

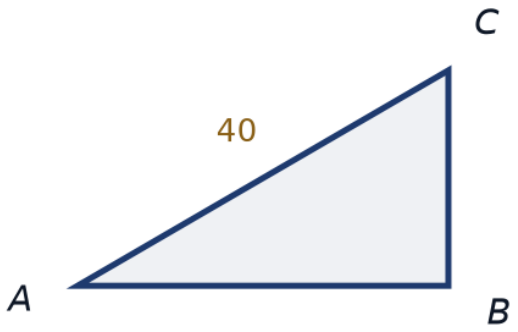
c. $\sqrt{1329}$

d. 51

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In a right triangle, one acute angle measures 30° and the hypotenuse is 40. Find the length of the side opposite the 30° angle, rounded to two decimal places.



a. 24.0

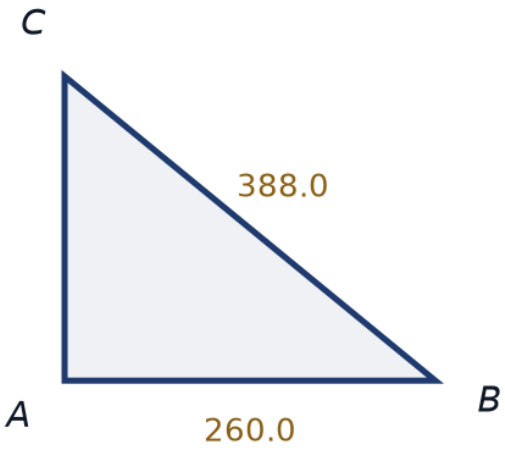
b. 18.0

c. 21.5

d. 20.0

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Circles

A right triangle has a hypotenuse of length 388 and one leg of length 260. What is the length of the other leg?



a. 388

b. 288

c. 260

d. 128

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45 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Area, Surface Area & Volume**

Two coastal observation stations, A and B, are situated 12 km apart along a straight shoreline running due east-west, with station B located due east of station A. Both stations detect a distressed vessel at point C offshore (north of the shoreline). From station A, the vessel is on a bearing of 030°. From station B, the vessel is on a bearing of 315°. What is the area of ΔABC, in square kilometers? *(Note: Bearings are measured clockwise from true north.)*

a. $108 - 36\sqrt{3}$

b. $108 + 36\sqrt{3}$

c. $36\sqrt{3} - 36$

d. $216 - 72\sqrt{3}$

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